to decide the best number of centroids to use with k-means.

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[1] That's also the reason why every company, online or offline, tries to discover as much information about you as possible: data is paramount in this era.

[2] Generative Adversarial Networks are a particular type of deep neural networks, where two competing models are trained to generate artificial content (based on a training) and to discriminate artificial from real content; their co-evolution increases the lifelikeness of the content generated.

[3] `NP-Hard` class of problems includes those problems that are at least as hard as the hardest problems in `NP`; as we have already seen in chapter 2, the `NP` class contains those problems that can be solved in polynomial time on a non-deterministic machine, and in particular the problems in `NP` but not in class `P` can't be solved in polynomial time on a deterministic machine. To date, determining if there is any problem in `NP-P` is the one of greatest challenges in computer science.

[4] For instance, in Python you can import `sample` from module `random`, or in JavaScript you can use underscore library's `sample` method.

[5] But if you do, then you should check out chapter 11 of *Practical Probabilistic Programming*, where it is neatly explained.

[6] In a `d`-dimensional space, two sets S_1 and S_2 are linearly separable if there exists at least one `(d-1)`-dimensional hyperplane that divides the space such that all points of S_1 are on one side of the hyperplane, and all points of S_2 are on the other side of it.

[7] As we have seen, k-means can discard some centroids, but this effect is very limited and unpredictable.

[8] Beyer, Kevin, et al. "When is "nearest neighbor" meaningful?" International conference on database theory. Springer, Berlin, Heidelberg, 1999.

[9] On GPUs and processors designed for vector computing, operations can be executed simultaneously on all coordinates of the `d`-dimensional tuples with higher efficiency.

[10] https://docs.scipy.org/doc/scipy-0.14.0/reference/generated/scipy.spatial.KDTree.html

[11] Ester, Martin, et al. "A density-based algorithm for discovering clusters in large spatial databases with noise." Kdd. Vol. 96. No. 34. 1996.
The approach used is closely related to another paper from 1972: Ling, Robert F. "On the theory and construction of k-clusters." The computer journal 15.4 (1972): 326-332.

[12] Technically, if two runs of DBSCAN cycle through the points in the same order, then the final result will be exactly the same. For k-means, instead, to be considered deterministic, we should replace the random initialization with a deterministic one.